Product Requirements Document

GBM2804H-100T Gimbal Motor Control

Overview

Develop firmware for the **NUCLEO-G431RB** with the **X-NUCLEO-IHM16M1** motor-driver board to operate the **GBM2804H-100T** three-phase gimbal motor using sensorless Field-Oriented Control (FOC).

Target Hardware

Item	Specification
MCU Board	NUCLEO-G431RB
Motor Driver	X-NUCLEO-IHM16M1 / STSPIN830
Motor	GBM2804H-100T, 7 pole pairs
Supply	12 V DC
Motor Connection	U, V, W three-phase output
Current Sensing	Three-shunt current sensing

Functional Requirements

1. Implement sensorless FOC for three-phase motor control.
2. Motor shall remain stopped after system initialization.
3. Support START and STOP commands through UART.
4. Support configurable speed command from 300 to 1000 RPM.
5. Support clockwise and counter-clockwise rotation.
6. Use hardware PWM generation for the three-phase inverter.
7. Measure motor phase current using the IHM16M1 current-sensing circuitry.
8. Limit commanded motor current to <= 1.5 A RMS or a lower validated project limit.
9. Detect motor-control faults and immediately disable PWM output.

UART Interface

UART: **115200 baud, 8-N-1**. Required commands:

MOTOR START	Start motor
MOTOR STOP	Stop motor
SPEED <RPM>	Set target speed
DIR CW / DIR CCW	Set direction
STATUS	Report state, target/estimated speed, direction and fault status

Validation Criteria

- Reliable motor startup from standstill
- Stable rotation at commanded speed
- Speed change while running
- CW and CCW operation
- Controlled motor stop
- Immediate PWM shutdown on motor-driver fault
- No operation outside configured voltage/current limits

Implementation Note: The G431RB + IHM16M1 + GBM2804H-100T combination is intended for three-phase BLDC/PMSM motor-control evaluation. Final speed, startup and current limits shall be tuned and validated on the actual motor and supply before extended operation.